

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK
Practical – VI

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6.Node.js and Express Fundamentals

a. Install Express and create a simple server.

```
PS R:\MERN STACK\Expresspractical> npm init -y
Wrote to R:\MERN STACK\Expresspractical\package.json:

{
  "name": "expresspractical",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
PS R:\MERN STACK\Expresspractical> npm install express

added 68 packages, and audited 69 packages in 2s

27 packages are looking for funding
run `npm fund` for details
```

```
PS R:\MERN STACK\Expresspractical> node server.js
Server is running on http://localhost:3000
```

b. Display a message on the browser using the server response.

Code :

```
const express = require("express");
```

```
const app = express();
```

```
const PORT = 3000;
```

```
app.get("/", (req, res) => {
  res.send("Welcome to Express Server");
});
```

```
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```



Welcome to Express Server

c. Create different routes in Express.

Code :

```
const express = require("express");
```

```
const app = express();
```

```
const PORT = 3000;
```

```
// Home Route
```

```
app.get("/", (req, res) => {  
  res.send(`  
    <h1>MVLU College Management Server</h1>  
    <p>Welcome to our college website.</p>  
  `);  
});
```

```
// Students Route
```

```
app.get("/students", (req, res) => {  
  res.send(`  
    <h1>Student List</h1>  
    <ul>  
      <li>Raj prajapati</li>  
      <li>Priya Patil</li>  
      <li>Amit Joshi</li>  
      <li>Neha Bhosale</li>  
    </ul>  
  `);  
});
```

```
// Faculty Route
```

```
app.get("/faculty", (req, res) => {
  res.send(`
    <h1>Faculty List</h1>
    <ul>
      <li>Sneha Gokarnkar - Information Technology</li>
      <li>Rohini Bhosale - Information Technology</li>
      <li>Sumit Tripathi - Information Technology</li>
      <li>Chetana Tanavade - Information Technology</li>
    </ul>
  `);
});

// Courses Route
app.get("/courses", (req, res) => {
  res.send(`
    <h1>Available Courses</h1>
    <ul>
      <li>B.Sc. Information Technology</li>
      <li>B.Sc. Computer Science</li>
      <li>B.Sc. Biotechnology</li>
      <li>B.A. Multimedia and Mass Communication</li>
    </ul>
  `);
});

// Contact Route
app.get("/contact", (req, res) => {
  res.send(`
    <h1>Contact Us</h1>
    <p>Email: info@mvlucollege.in</p>
    <p>Phone: 9876543210</p>
    <p>Address: Andheri, Mumbai</p>
  `);
});

// Start Server
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
Output :
```



MVLU College Management Server

Welcome to our college website.

Student List

- Ashwin Tripathi
- Priya Patil
- Amit Joshi
- Neha Bhosale

d. Send a JSON response from the server.

Code :

```
const express = require("express");
```

```
const app = express();
```

```
const PORT = 3000;
```

```
app.get("/", (req, res) => {  
  res.send("Welcome to MVLU College Management Server");  
});
```

```
app.get("/students", (req, res) => {
```

```
  const students = [  
    {  
      id: 1,  
      name: "Ashwin Tripathi",  
      course: "B.Sc. IT",  
      marks: 85  
    },  
    {  
      id: 2,  
      name: "Priya Patil",  
      course: "B.Sc. IT",  
      marks: 92  
    },  
    {  
      id: 3,  
      name: "Amit Joshi",  
      course: "B.Sc. CS",  
      marks: 78  
    }  
  ]  
}
```

```

    }
  ];

  res.json(students);
});

app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});

```

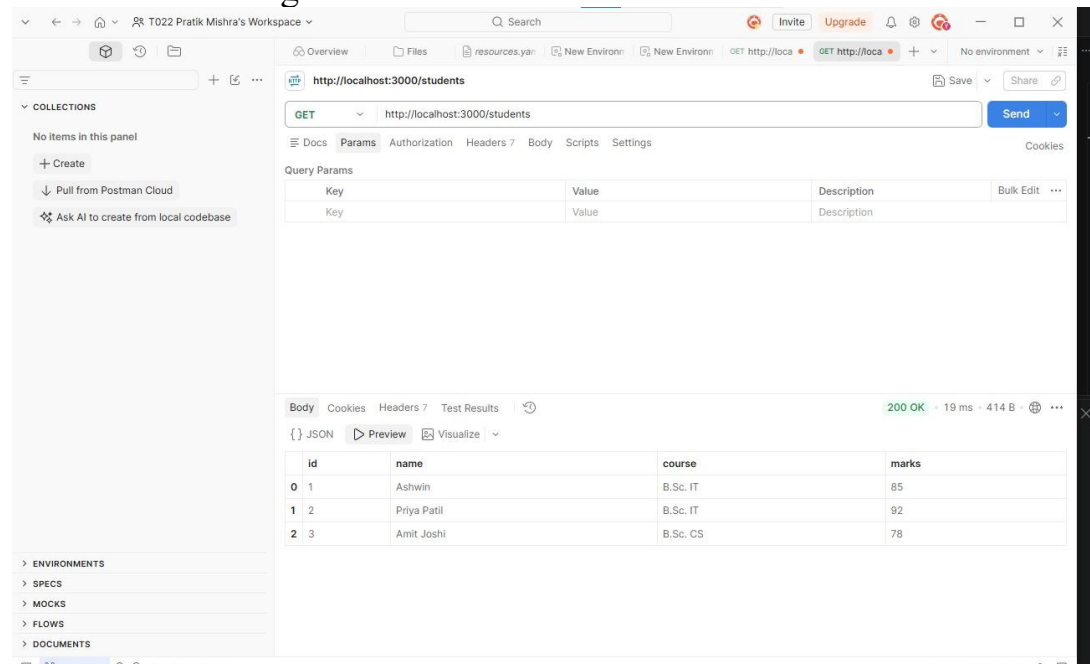
Output :

```

[
  {
    "id": 1,
    "name": "Ashwin Tripathi",
    "course": "B.Sc. IT",
    "marks": 85
  },
  {
    "id": 2,
    "name": "Priya Patil",
    "course": "B.Sc. IT",
    "marks": 92
  },
  {
    "id": 3,
    "name": "Amit Joshi",
    "course": "B.Sc. CS",
    "marks": 78
  }
]

```

e. Test routes using browser or Postman.



The screenshot shows the Postman interface with a GET request to `http://localhost:3000/students`. The response is a 200 OK status with a JSON body containing an array of student objects.

id	name	course	marks
0 1	Ashwin	B.Sc. IT	85
1 2	Priya Patil	B.Sc. IT	92
2 3	Amit Joshi	B.Sc. CS	78

f. Run the server and observe the output in the console.

```

PS R:\MERN STACK\Expresspractical> node server.js
Server is running on http://localhost:3000
PS R:\MERN STACK\Expresspractical> node --watch server.js
Server is running on http://localhost:3000

```